MIL-M-46039B(MR)
23 January 1987
SUPERSEDING
MIL-M-46039A(MR)
1 October 1965

MILITARY SPECIFICATION

MAGNESIUM ALLOY BAR, ROD, AND SPECIAL SHAPED SECTIONS, EXTRUDED (ZK21A)

This specification is approved for use by the U.S. Army Materials Technology Laboratory, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers magnesium alloy bar, rod, and special shaped sections in the "as extruded" temper (ZK21A).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-3171 - Magnesium Alloy; Process for Corrosion Protection of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Laboratory Command, Materials Technology Laboratory, ATTN: SLCMT-MEE, Watertown, MA 02172-0001 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

/FSC 9530/

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STANDARDS

FEDERAL

FED-STD-184 - Identification Marking of Aluminum, Magnesium and Titanium
FED-STD-245 - Tolerances for Aluminum Alloy and Magnesium Alloy Wrought Products

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-649 - Aluminum and Magnesium Products, Preparation for Shipment and Storage

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

ASTM

B107 - Magnesium - Alloy extruded Bars, Rods Shapes, Tubes and Wire.
E8 - Tension Testing of Metallic Numerals
E35 - Chemical Analysis of Magnesium and Magnesium Alloys
E55 - Sampling Wrought Nonferrous Metals and Alloys For Determination of Chemical Composition

(Application for copies shaould be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Chemical composition. The chemical composition shall conform to the requirements shown in Table I.

TABLE I. Chemical composition.

Element	Percent
Zinc	2.0 -2.6
Zirconium	0.45-0.8
Magnesium	Remainder
Total other elements, max.	0.30[1/]

1/ Analysis shall regularly be made only for the elements specifically mentioned in the table. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine that the total thereof does not exceed 0.30 percent.

3.1.1 The contractor shall furnish an analysis of each melt showing the percentage of each of the elements designated in table I.

3.2 Mechanical properties. The mechanical properties of the extruded bar, rod, and shapes shall be as specified in table II.

TABLE II. Mechanical properties requirements.

Form	Cross-section area, sq. in.	Tensile strength, minimum, psi	Yield strength, minimum, psi	Elongation in 2 inches, minimum, percent
Bar, rod, and shapes	Under 5.000	38,000	28,000[1/]	4

1/ Yield strength at 0.2 percent offset or at 0.0126 inch in 2 inches extension under load.

3.3 Dimensional tolerances. The extruded bar, rod, and shapes shall not vary from the specified dimensions, straightness, contour, twist, flatness, leg angles and corner and fillet radii by amounts greater than the applicable tolerances specified in FED-STD-245 for cross-sectional dimensions, and straightness, twist, contour and surface roughness of extruded bar, rod, and shapes; for squareness of cut for extruded and rolled bar, rod, and shapes; for flatness and corner and fillet radii of extruded bar and shapes; and for leg angle of extruded shapes.

3.4 Marking for identification. Unless otherwise specified in the contract or order, marking shall be in accordance with FED-STD-184. In addition to the commercial alloy designation, the number of this specification shall be applied to each extrusion.

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3.5 Requirements for sizes not specifically covered. Mechanical properties for sizes of material not covered by this specification shall be as specified in the contract or order.

3.6 Protective treatment. The bar, rod, and shapes shall be either oiled or chrome-pickled as a final process before shipment. Unless otherwise specified (see 6.2), the extrusions shall be given a chrome-pickle treatment in accordance with type I of MIL-M-3171.

3.7 Workmanship. The extruded bar, rod, and shapes shall be uniform in quality and condition; clean, sound, smooth, and free from seams, slivers, laminations, grooves, cracks, and other injurious defects. Discoloration due to heat treatment shall not be cause for rejection.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Lot. A lot shall consist of material of the same section, size, and temper submitted for inspection at one time.

4.3 Sampling.

4.3.1 For chemical analysis. A sample for chemical analysis shall be obtained from each 4,000 pounds or less of material comprising the lot, and analyzed in accordance with 4.5.2.1. When the sample is taken from the fabricated product, it shall consist of equal quantities of chips taken from at least four representative pieces and shall represent the full thickness of the material. Upon approval of the procuring activity, the manufacturer's method of composition control may be used in lieu of the methods outlined above.

4.3.2 Mechanical properties. From material having a nominal weight of less than one pound per lineal foot, one tension test sample shall be selected from each lot weighing 500 pounds or less; from lots greater than 500 pounds, one additional sample shall be selected for each 1,000 pounds or fraction thereof

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in excess of the first 500 pounds. From material having a nominal weight of one pound or more per lineal foot, one tension test sample shall be selected from each lot consisting of 500 feet or less; from lot consisting of more than 500 feet, an additional sample shall be selected for each 1,000 feet or fraction thereof in excess of the first 500 feet.

4.4 Examination. Each bar, rod, and shape shall be carefully examined to determine conformance with the specification with respect to workmanship. Sufficient spot checks shall be made to insure conformance with the tolerance specified. A system of statistical quality control may be used for dimensional and workmanship examination.

4.4.1 Preparation for shipment. Examination of the preservation, packaging, packing and marking for shipment shall be made for conformance to the requirements of section 5.

4.5 Tests.

4.5.1 Tension test specimens. Tension test specimens shall conform to the requirements of ASTM E8. Whenever applicable, the material may be tested in full section. For material less than 1/2 inch in section thickness and of suitable width, which is not tested in full section, a test specimen conforming to Figure 6 (ASTM E8), sheet type, 1/2 inch wide, shall be used. For material 1/2 inch or more in section thickness, which is not tested in full section, a test specimen conforming to the appropriate part of Figure 8 (ASTM E8) shall be used. For shapes from which a standard test specimen cannot be taken and which cannot be tested satisfactorily in full section, a strip taken from the section shall be tested.

4.5.2 Test procedures.

4.5.2.1 Chemical analysis. The samples selected and forwarded in accordance with 4.3.1 shall be analyzed in accordance with the provisions of ASTM E35.

4.5.2.2 Tension tests. All tension tests shall be in accordance with the requirements of ASTM E8.

4.6 Rejection.

4.6.1 Examination defects. Any sample unit having one or more defects shall be rejected.

4.6.2 Test failures. A lot shall be rejected for failure to meet any of the test requirements when tested in accordance with 4.5 subject to the provisions of 4.6.2.1 and 4.6.2.2.

4.6.2.1 Rejection. When one or more test specimens fail to meet the requirements of the specification, the lot represented by the specimen or specimens shall be subject to rejection.

4.6.2.2 Retest. When no sampling plan is provided by the procuring agency and where there is evidence that indicates that the specimen was not representative of the lot material, and when the detail specification does not

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otherwise specify, at least two specimens shall be selected to replace each test specimen which failed. All specimens so selected for retest shall meet the requirements of the specification or the lot shall be subject to rejection.

5. PACKAGING

5.1 Preservation. Unless otherwise specified in the contract or order (see 6.2), all material shall be preserved in accordance with MIL-STD-649C.

5.1 Packaging. Unless otherwise specified in the contract or order, all material shall be separated by temper, size, and thickness and packed in crates or boxes of the size commonly used for shipment. When specified (see 6.2), individual extruded bar, rod, and special shaped sections which have been given a chrome pickle surface treatment shall be separated by a non-corrosive, non-hygroscopic paper to protect the finish. Oiled bar, rod, or special shaped sections shall not be required to be separated by paper.

5.2.1 Level A. The bar, rod, and special shaped sections, shall be packed for shipment in containers of a type and size normally used for the purpose and in a manner to insure carrier acceptance and safe delivery to destination at the lowest applicable rate.

5.2.2 Levels B and C. Bar, rod, and special shaped sections, extruded, shall be prepared for shipment as described in MIL-STD-649.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. ZK21A alloy has good extrusion characteristics and moderate mechanical properties. It is readily welded and does not require a stress relieving treatment after welding.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Temper of material required.
(c) Length, size (die number and shapes).
(d) Mechanical properties for sizes or tempers not covered by Table II.
(e) Mechanical properties and tolerances for material having dimensions outside the specification limits.
(f) Purpose for which bar, rod, or special shapes, extruded, is intended.
(g) Whether chrome pickled or oiled bar, rod, or special shapes, extruded, is required (see 3.6).
(h) Whether material is to be packaged by level A, B or C (see 5.1.1 and 5.1.2).
(i) Special marking, if required (see 3.4 and 5.2).
(j) Interleave packaging, if required (see 5.1).

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6.3 The requirements for packaging (see 5.1) and marking for shipment (see 5.2) specified herein apply to direct shipments for Government activities and apply also, where specified, to contracts or orders between the manufacturer and the Government prime contractor.

6.4 Subject term (key word) listing.

Bar
Extruded
Magnesium alloy
Rod
Special shaped sections
ZK21A

6.5 Metric conversion.

METRIC UNITS AND CONVERSION FACTORS

Multiply	By	To obtain
Feet	0.3048	Metres
Pounds	0.4536	Kilograms
Square inches	6.452	Square centimetres
Psi	6.8948	KPa

Custodian:
Army - MR

Review interest:
Army - AR

User interest:
Army - ME

Preparing activity:
Army - MR

Project 9530-A012

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-46039B(MR)	MAGNESIUM ALLOY BAR, ROD, AND SPECIAL SHAPED SECTIONS, EXTRUDED (ZK21A)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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